

Matter Library Project — Executive Summary

Unified Physically-Based Materials Across Blender 5.1+ and Unreal Engine (Substrate)

The **Matter Library Project** establishes a single, physically grounded material ecosystem that delivers consistent visual results across real-time and offline rendering. It replaces traditional texture fakery with real-scale, physically correct materials built from the same canonical source: **MaterialX**. Our goal is to enable **lifelike, real-world materials**—from stone and metal to glass, fabrics, and foliage—that behave predictably whether viewed in Blender, Unreal Engine, or other physically-based renderers. The result is a shared material foundation for visualization, design, and simulation. **Architecture**

MaterialX serves as the open, single source of truth. Each material (a “Matter”) defines physical parameters, overlays, and masks within standardized nodegraphs. Bridges convert these into Blender’s Principled BSDF and Unreal Engine’s Substrate materials, ensuring **visual parity** between offline and real-time workflows. Runtime tools dynamically modify parameters such as color, roughness, and scale without reimporting or recompiling.

Standardization

All 145 base materials follow a unified taxonomy across 38 Families (stone, metal, plastic, etc.), grouped by five *MetaOrigins*: Natural, Synthetic, Engineered, Environmental, and Utility. Each uses one of five **Master Materials**—Opaque, Masked, TranslucentThin, TwoLayer, and Emissive—sharing a common parameter set verified across Blender and Unreal. **Key Features**

- Real-world scale (defined meters-per-tile)
- Reusable RGBA Overlays and MaskSets
- Deterministic naming and metadata
- 8K physically accurate textures (scalable)
- Lumen-ready Substrate support
- Fully open MaterialX-based schema (ASWF license)
- MaterialX schema v2 (Overlay & Mask nodes)
- Blender and Unreal master material templates
- Automated MaterialX → UE/Blender bridges
- Parity QA testing ($\Delta E < 2$)
- Open-source release and contributor guidelines

Outcome

The Matter Library provides a future-proof, open framework for physically based materials that look and behave the same everywhere. By aligning with MaterialX, Blender, and Unreal’s Substrate, it enables a unified approach to realism, collaboration, and cross-platform material fidelity.